

For a bank to properly determine which methods are to be employed, data science techniques such as hypothesis testing, crowdsourcing, machine learning, natural language processing and visualization can be applied to increase throughput and discover insights that were never thought to have previously existed.



Banking on Big Data

Fraud detection

In the current context, the areas within the banking industry that present the strongest near-term opportunities for tangible performance improvement include risk management/fraud detection and improved customer communication and loyalty.

One of the main priorities of the banking sector,

especially after the financial crisis of 2008, has been timely fraud detection, with one the earliest examples being the FICO Falcon fraud assessment system, which operated under a neural network shell. Most premium software available in the market have evolved and measure up to current standards, by leveraging the power of state-of-the-art big data techniques to capture information available from the copious amounts of data. Such systems usually look for anomalies buried in the spending patterns of a customer. These patterns are usually captured in from their transaction data, which contains the location, amount and time of said transaction, to detect whether any sort of credit card fraud or identity theft has been committed. Such methods can also minimise cases of false alarms being raised (false positive), and help smoothen the customer experience.

Customer Satisfaction

A different, but popular use is in risk and credit profiling. When a customer applies for a loan in a bank, banks have traditionally been known to take days to process the credit worthiness of a customer before replying in the affirmative or negative for the loan application. In order to better the cus-